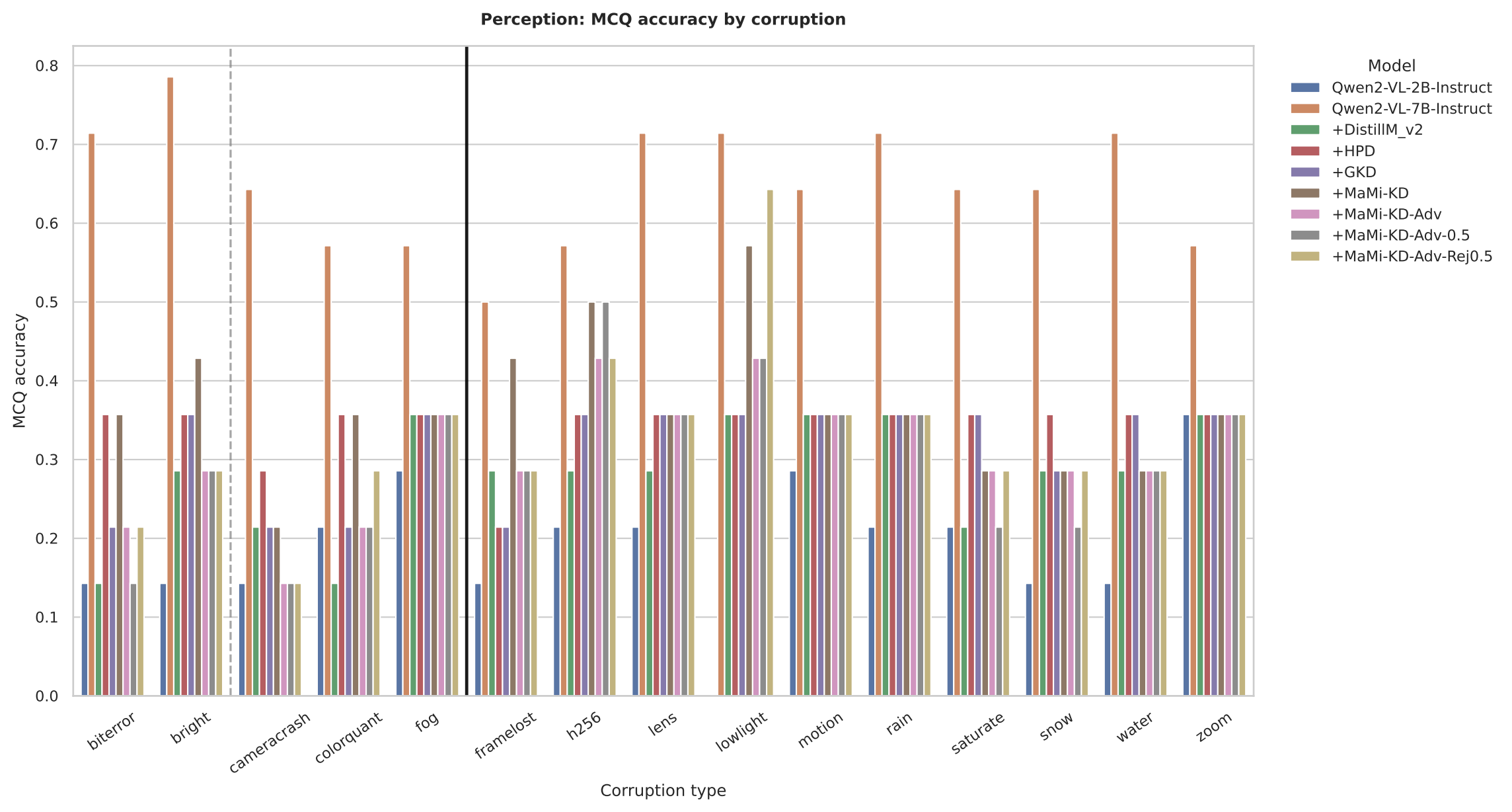
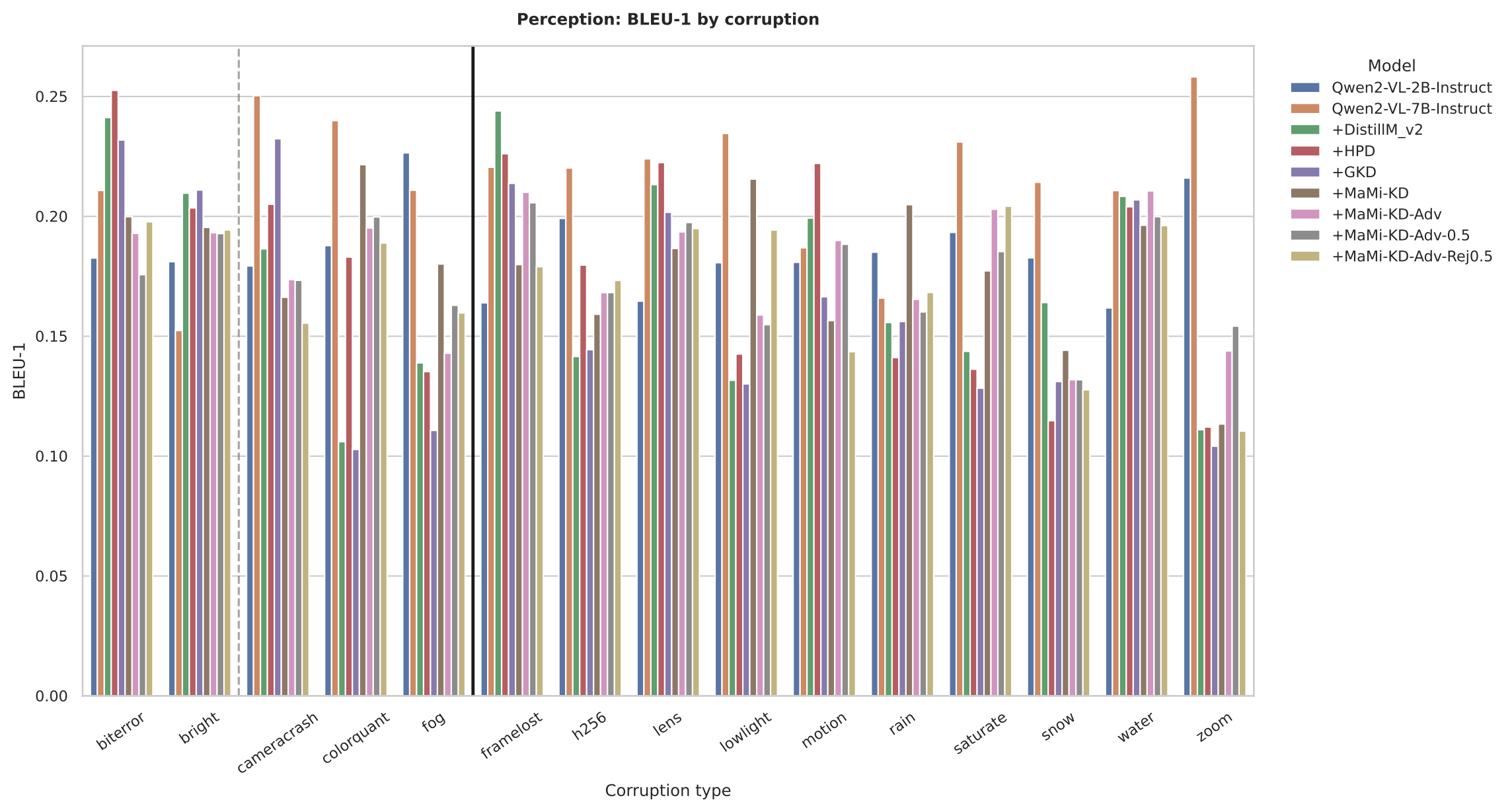
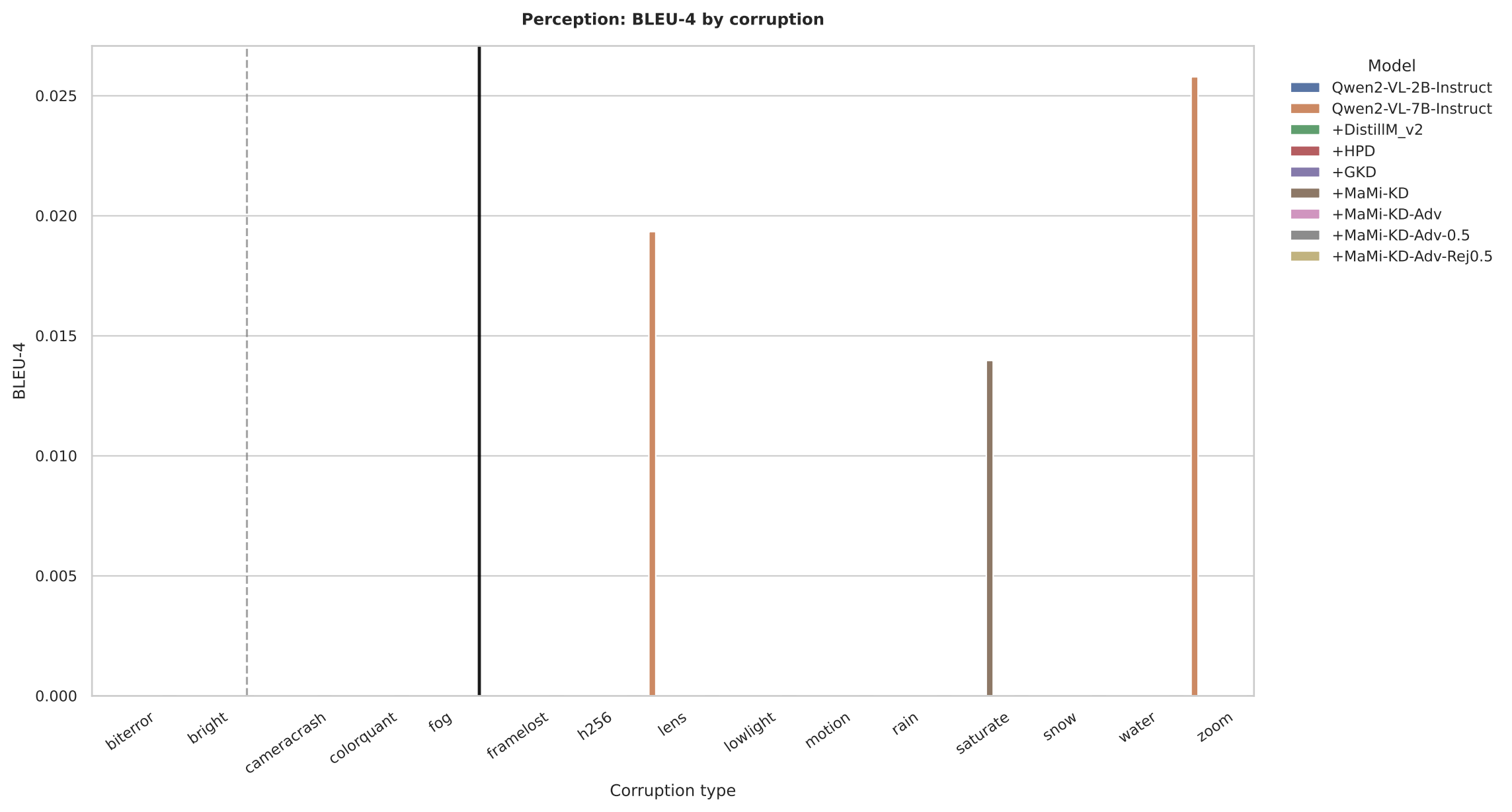
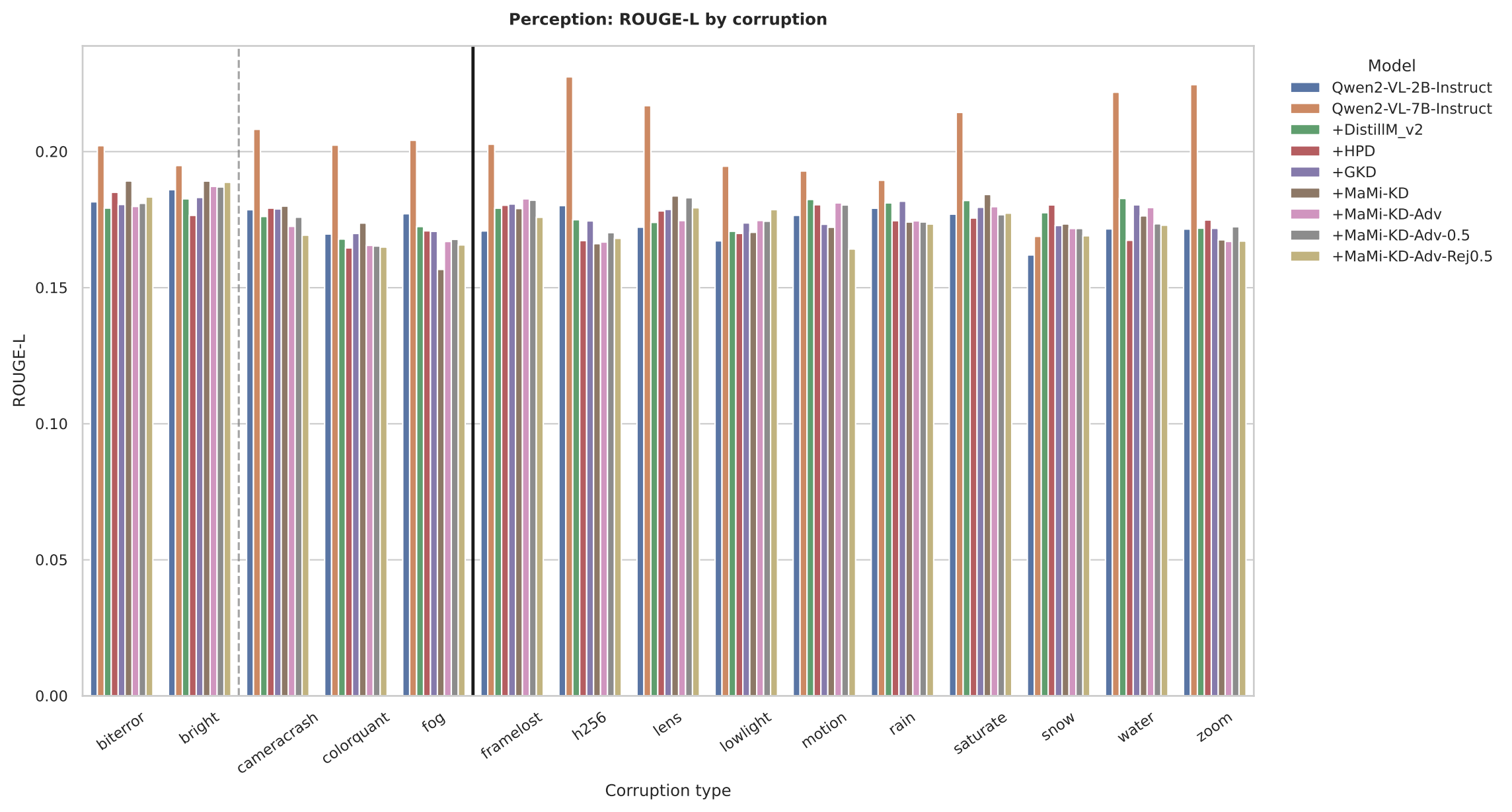
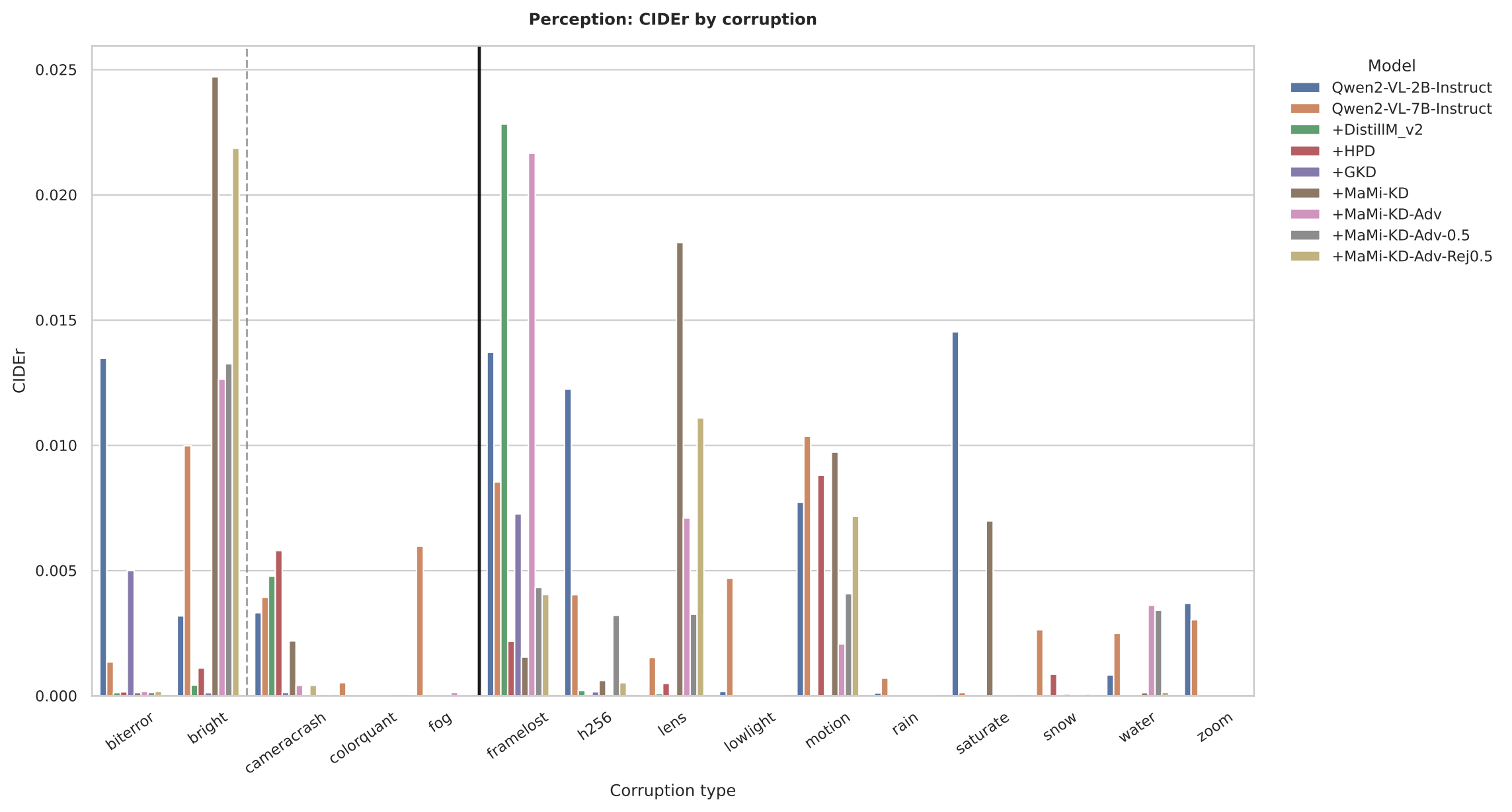
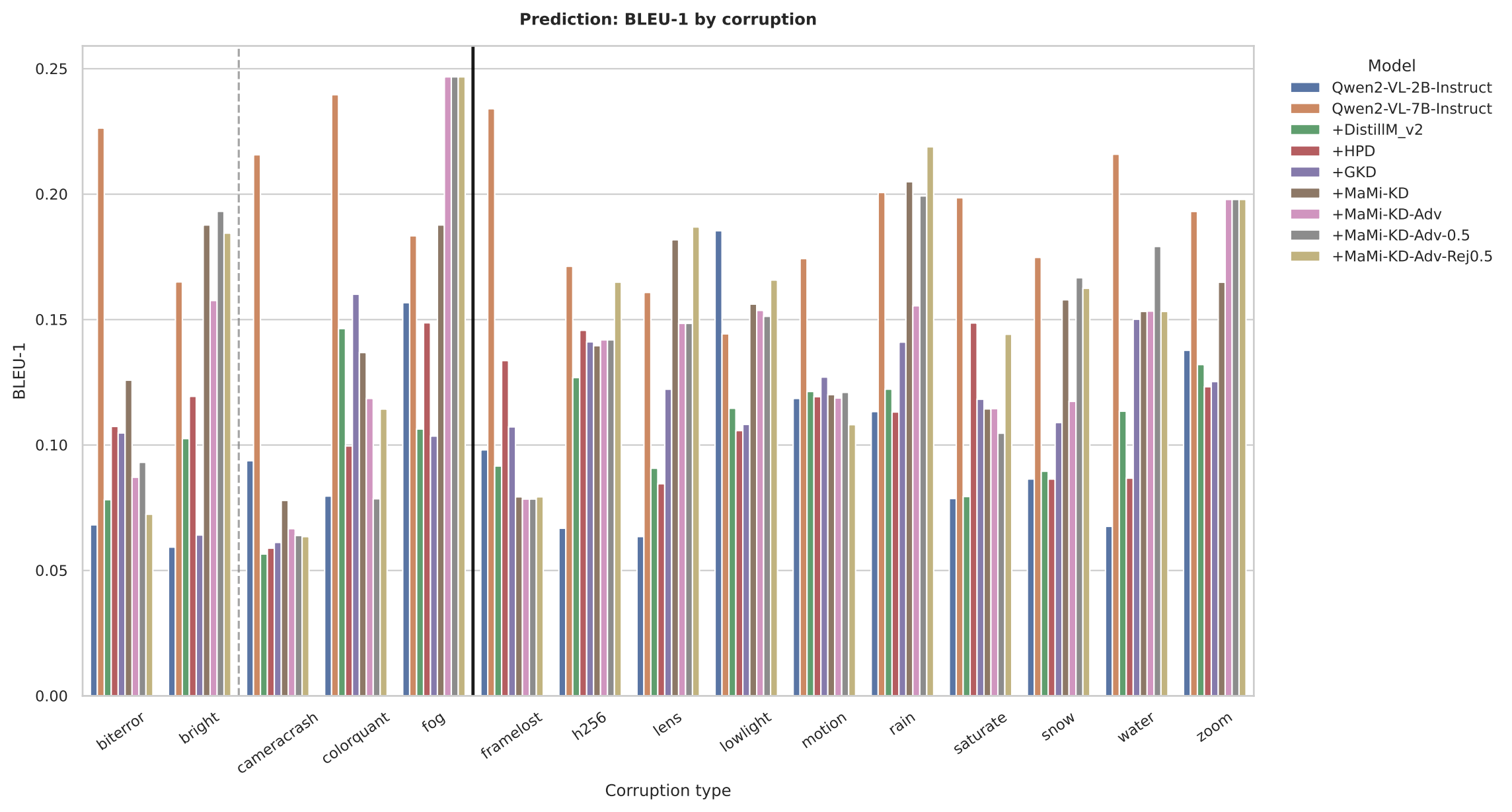

Perception: MCQ accuracy by corruption
MCQ accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

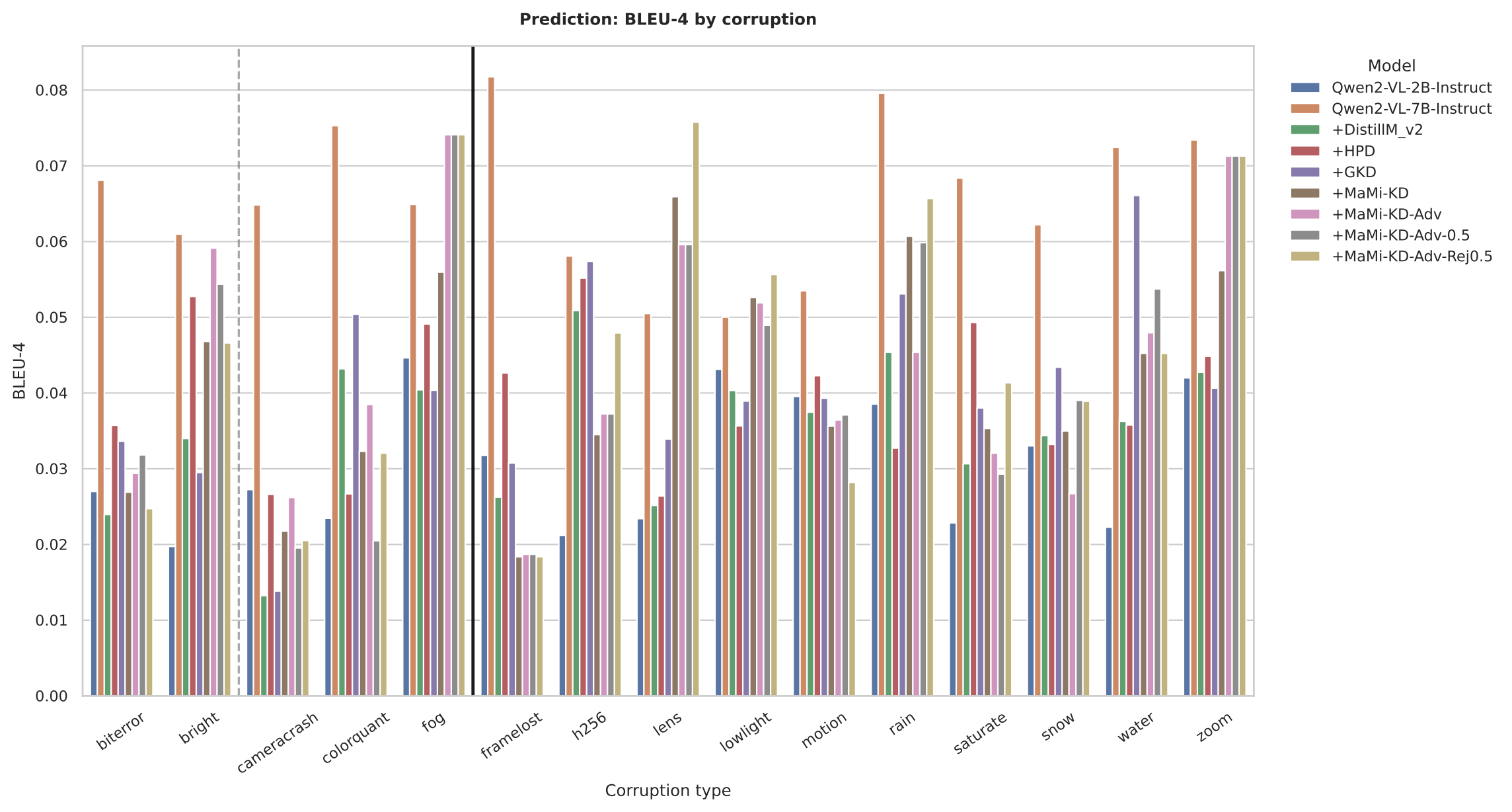

Perception: BLEU-1 by corruption
BLEU-1
0.00
0.05
0.10
0.15
0.20
0.25
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

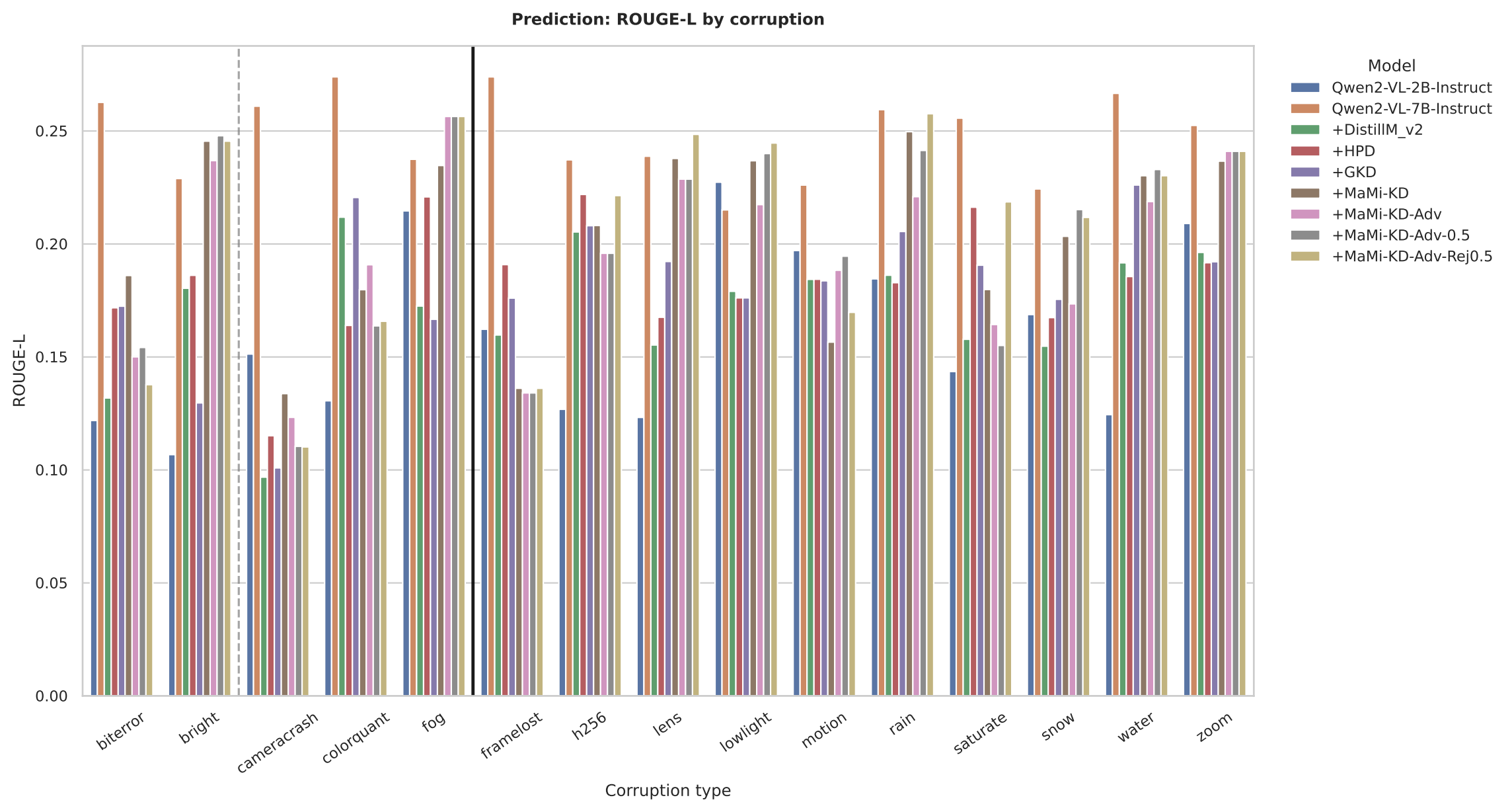

Perception: BLEU-4 by corruption
BLEU-4
0.000
0.005
0.010
0.015
0.020
0.025
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

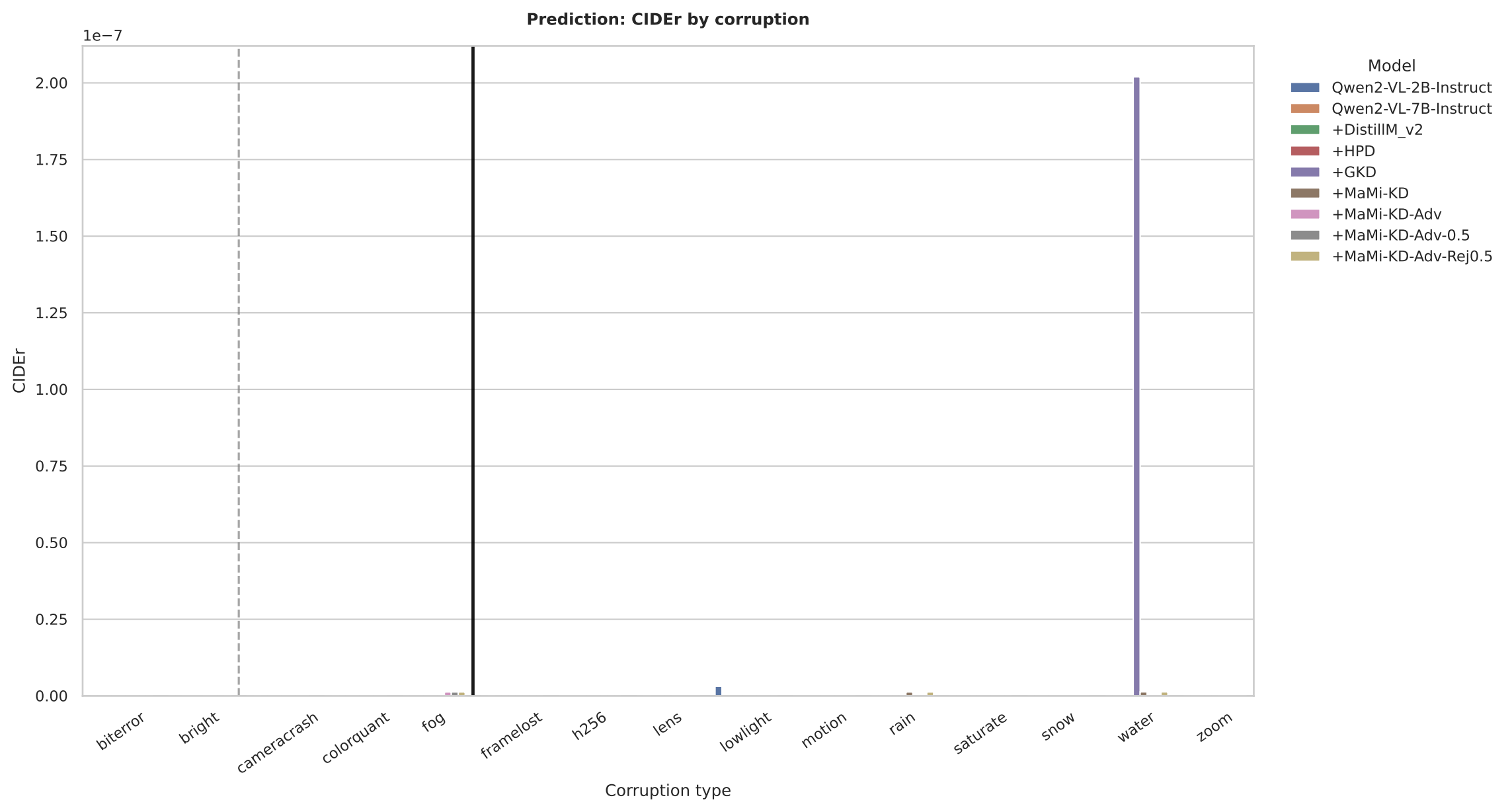

Perception: ROUGE-L by corruption
ROUGE-L
0.00
0.05
0.10
0.15
0.20
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

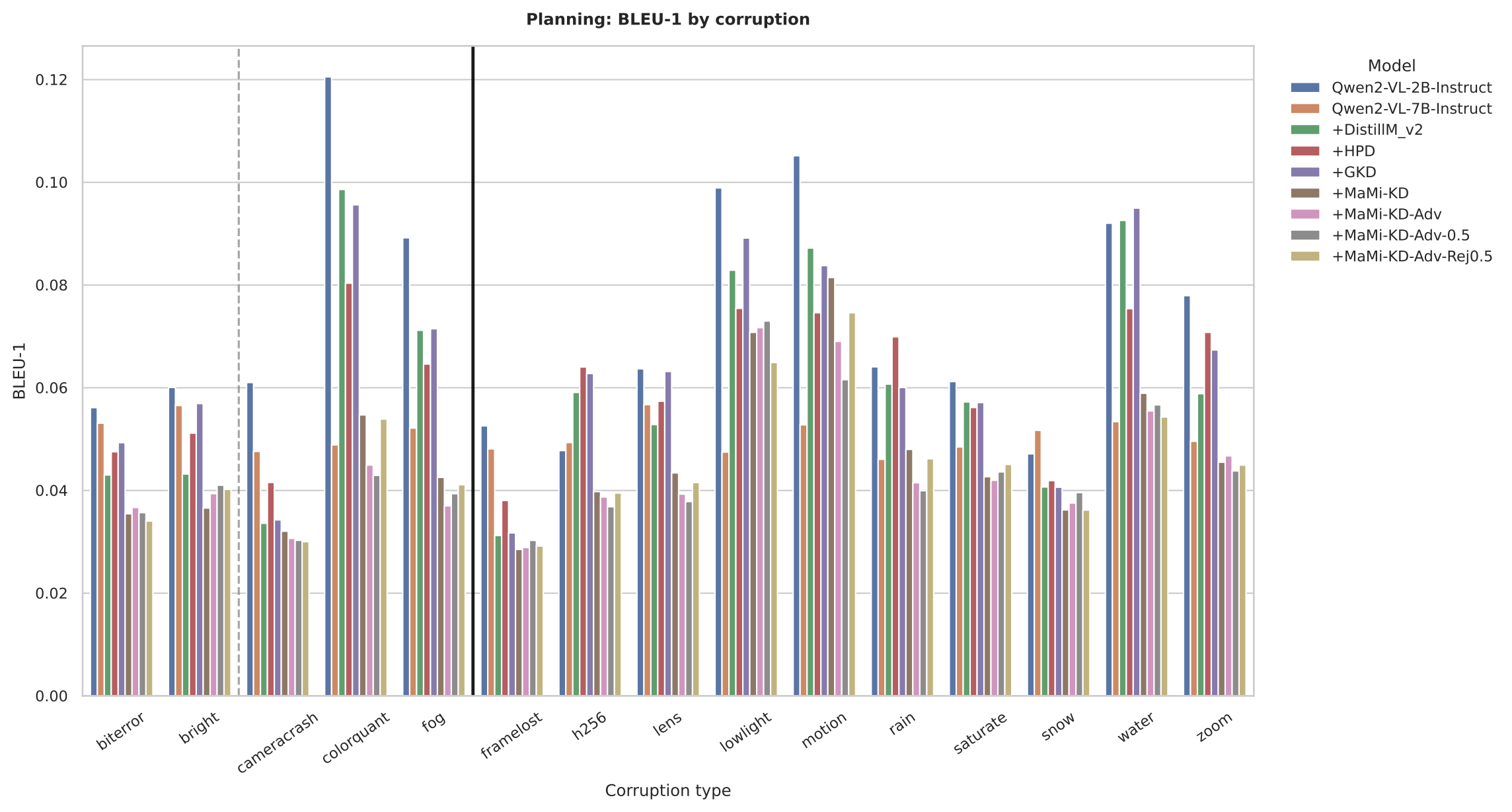

Perception: CIDEr by corruption
CIDEr
0.025
0.020
0.015
0.010
0.005
0.000
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

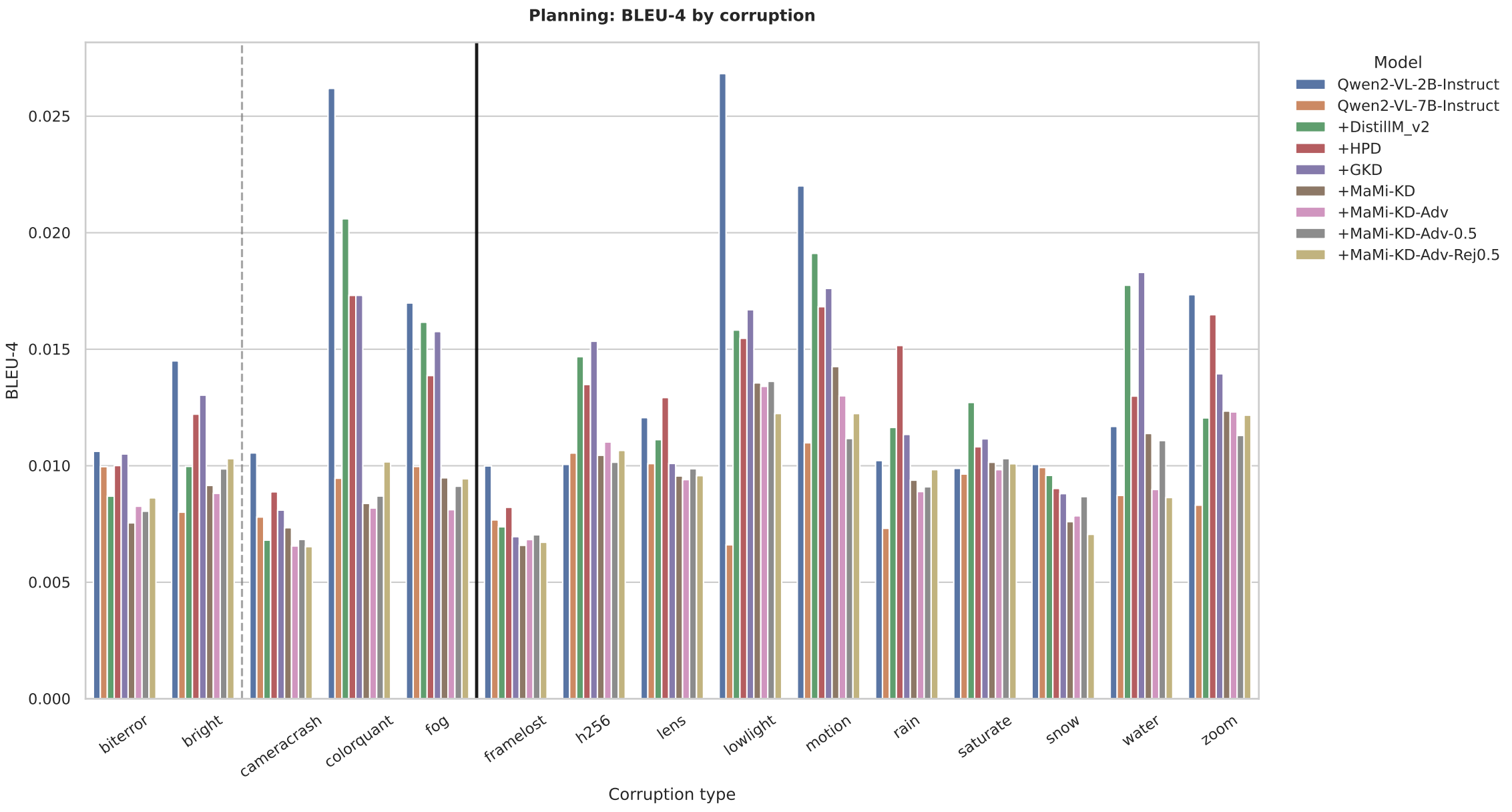

Prediction: BLEU-1 by corruption
BLEU-1
0.00
0.05
0.10
0.15
0.20
0.25
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

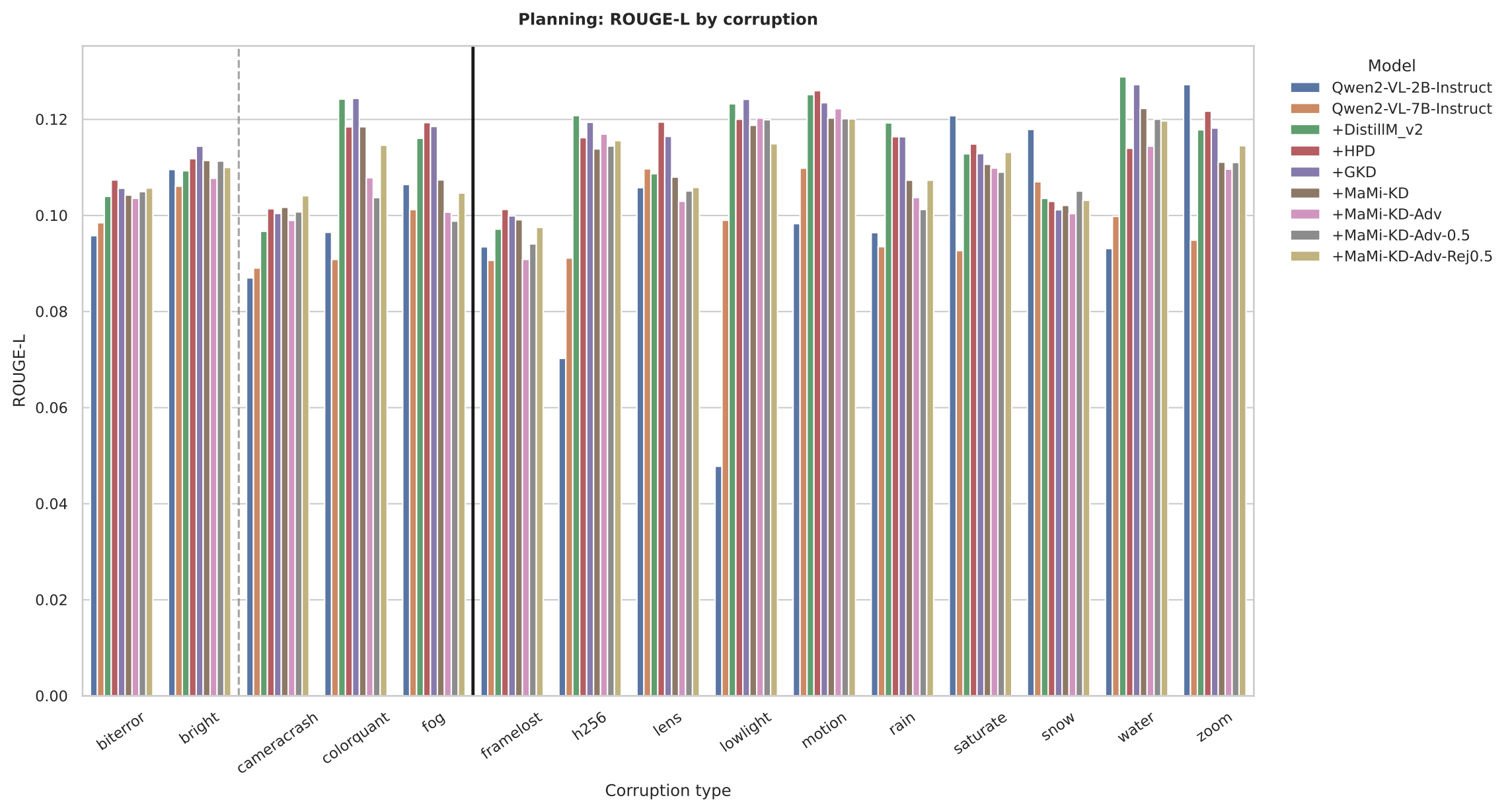

Prediction: BLEU-4 by corruption
BLEU-4
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

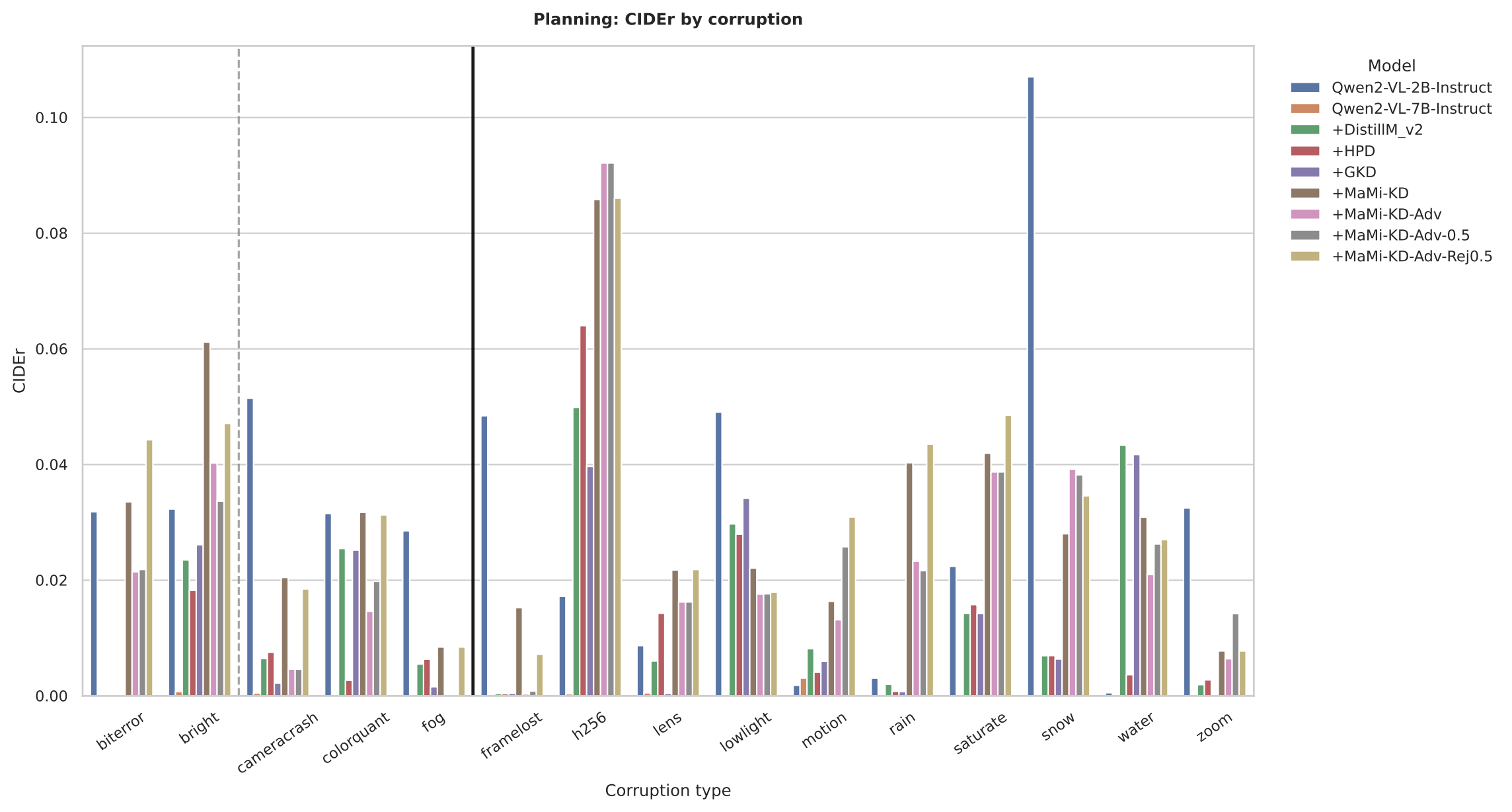

Prediction: ROUGE-L by corruption
ROUGE-L
0.00
0.05
0.10
0.15
0.20
0.25
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5

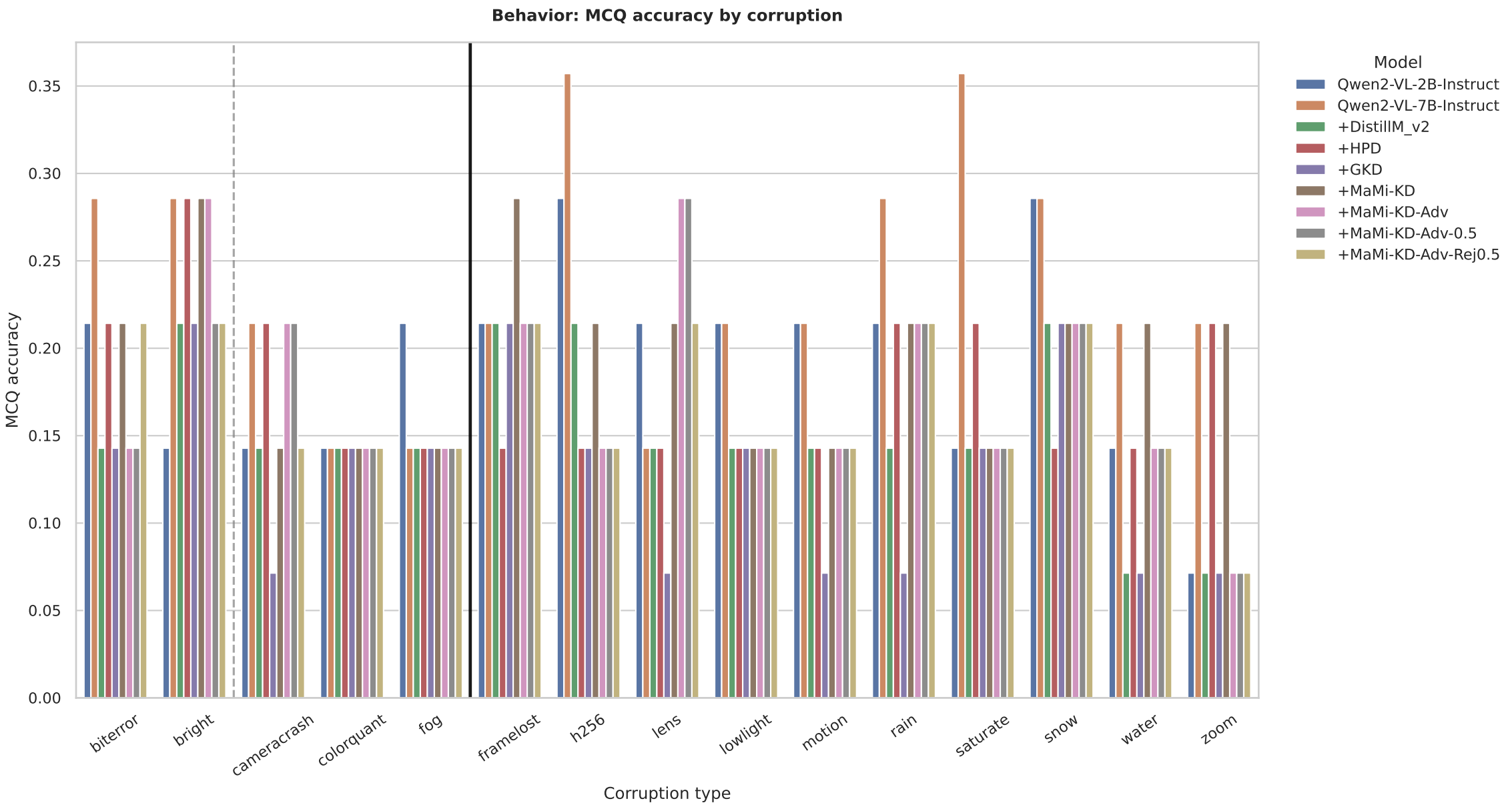

Prediction: CIDEr by corruption
1e−7
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
CIDEr
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5


Planning: BLEU-1 by corruption
BLEU-1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5


Planning: BLEU-4 by corruption
BLEU-4
0.000
0.005
0.010
0.015
0.020
0.025
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5


Planning: ROUGE-L by corruption
ROUGE-L
0.00
0.02
0.04
0.06
0.08
0.10
0.12
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5


Planning: CIDEr by corruption
CIDEr
0.00
0.02
0.04
0.06
0.08
0.10
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5


Behavior: MCQ accuracy by corruption
MCQ accuracy
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen2-VL-2B-Instruct
Qwen2-VL-7B-Instruct
+DistillM_v2
+HPD
+GKD
+MaMi-KD
+MaMi-KD-Adv
+MaMi-KD-Adv-0.5
+MaMi-KD-Adv-Rej0.5